Exploratory Data Analysis (EDA)

A Beginner-Friendly Guide — Theory & Simple Examples

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What you will learn

What EDA is and why it matters · The main steps of EDA · Summary statistics in plain language · How to find missing values, duplicates and outliers · The most popular charts and when to use each · Common mistakes to avoid

1. What is EDA?

Exploratory Data Analysis (EDA) is the first look we take at a set of data **before** drawing any conclusions or building any model. The goal is to **understand** the data: what it contains, whether it is clean, what patterns it shows, and what looks strange.

The term was made popular by the American statistician **John Tukey** in the 1970s. He encouraged people to look at data with open eyes instead of jumping straight to formulas.

> **Everyday analogy**
> A doctor does not prescribe medicine the moment you walk in. First they check your temperature, blood pressure and ask questions. EDA is that **check-up** for data. Only after it do we decide on the 'treatment' (the analysis or model).

Why is EDA important?

- **Catch errors early** – a student's age recorded as 210 or a salary typed with an extra zero.
- **Understand patterns** – e.g. sales always jump before Eid.
- **Ask better questions** – data often reveals questions you did not think of.
- **Choose the right method** – the shape of the data guides the later analysis.
- **Avoid wrong decisions** – 'garbage in, garbage out': bad data gives bad conclusions.

2. Basic Vocabulary

Most data is arranged like a table (an Excel sheet). Consider this small example:

Student	Department	Gender	Study hours/day	Marks
Rahim	Business	Male	3	55
Karima	Arts	Female	5	68
Sadia	Science	Female	(blank)	72
Tanvir	Business	Male	1	40

- **Dataset** – the whole table.
- **Observation / Record / Row** – one item being studied (one student).
- **Variable / Feature / Column** – one characteristic measured (Marks, Gender).
- **Missing value** – an empty cell, like Sadia's study hours.

Types of variables

Type	Meaning	Examples
Numerical – continuous	Measured; can take any value in a range	Height, weight, income, temperature
Numerical – discrete	Counted; whole numbers	Number of children, number of loans
Categorical – nominal	Groups with no natural order	Gender, department, district
Categorical – ordinal	Groups with a natural order	Grade (A, B, C), satisfaction (low/medium/high)

> **Why this matters**
> The type of a variable decides which summary and which chart you should use. You can calculate an average of marks, but an 'average department' makes no sense – for departments you count instead.

3. The Steps of EDA

Step	Question we ask	Example
1. Understand the data	What does each column mean? How many rows?	500 customers, 8 columns
2. Check data quality	Are there blanks, duplicates, typing errors?	12 customers have no age
3. Summarise	What is typical? How spread out?	Average age is 34
4. Univariate analysis	How does ONE variable behave?	Most customers are 25–40
5. Bivariate analysis	How do TWO variables relate?	Older customers spend more
6. Multivariate analysis	How do MANY variables interact?	Age, income and city together
7. Note insights	What did we learn? What next?	Target 25–40 age group

4. Summary Statistics (Describing Data with Numbers)

Example: marks of 8 students — **40, 50, 55, 60, 60, 65, 70, 80**

4.1 Measures of centre – 'What is typical?'

Measure	How to find it	Our example
Mean (average)	Add all values, divide by how many	(40+50+55+60+60+65+70+80) / 8 = 480 / 8 = **60**
Median	Middle value after sorting (average of the two middle ones if count is even)	Middle two are 60 and 60, so median = **60**
Mode	Most frequent value	60 appears twice, so mode = **60**

Mean vs. Median – a powerful lesson
Monthly salaries (thousand BDT) of 7 people in an office: 20, 21, 22, 23, 24, 25 and the owner's **200**.
Mean = 335 / 7 ≈ **47.9** Median = **23**
The mean suggests people earn about 48 thousand, but six of the seven earn 25 or less! One extreme value pulls the mean. **When data has extreme values, the median describes the 'typical' person better.**

4.2 Measures of spread – 'How different are the values?'

Measure	Meaning	Our example
Minimum / Maximum	Smallest and largest values	40 and 80
Range	Maximum – Minimum	80 – 40 = **40**
Quartiles (Q1, Q3)	Values that cut the sorted data into four equal parts	Q1 = 52.5, Q3 = 67.5
IQR (Interquartile range)	Q3 – Q1; spread of the middle 50%	67.5 – 52.5 = **15**
Standard deviation	Roughly, the average distance of values from the mean	Small = values close together; large = widely scattered

Standard deviation – intuition only
Class A marks: 58, 59, 60, 61, 62. Class B marks: 30, 45, 60, 75, 90. Both classes have a mean of 60, but Class B is far more spread out, so Class B has a much **larger** standard deviation. Same average, very different classes!

4.3 Shape – skewness

- **Symmetric** – both sides look similar; mean ≈ median (e.g. heights of adults).
- **Right (positive) skew** – a long tail of large values; mean > median (e.g. incomes, house prices).
- **Left (negative) skew** – a long tail of small values; mean < median (e.g. marks in a very easy exam).

4.4 Summaries for categorical data

For categories we use **frequency** (count) and **percentage**. Example: 125 students — Business 45 (36%), Arts 30 (24%), Science 25 (20%), Law 15 (12%), Others 10 (8%). The mode here is 'Business'.

5. Checking Data Quality

5.1 Missing values

Blank cells happen when people skip a survey question, a machine fails, or data is not entered. First ask: **how many** are missing and **why**?

Option	When it is reasonable	Example
Remove the rows	Very few rows are missing	3 out of 1,000 customers have no age
Fill with mean / median	Numerical column, moderate missing	Fill missing age with median age 34
Fill with mode	Categorical column	Fill missing city with 'Dhaka' (most common)
Keep as 'Unknown'	The blank itself may carry meaning	'Did not disclose income'
Drop the column	Most of the column is empty	70% of 'fax number' is blank

5.2 Duplicates

The same record entered twice. If customer 'Nusrat, 01711-XXXXXX' appears twice, her purchases are counted double and total sales look bigger than reality. Duplicates should be found and removed.

5.3 Inconsistent entries and impossible values

- Same thing written differently: 'Dhaka', 'dhaka', 'DHAKA', 'Dhk' – these should become one category.
- Impossible values: age = –5, marks = 150 out of 100, a date of birth in the future.
- Wrong units: some weights in kg and some in pounds in the same column.

5.4 Outliers

An **outlier** is a value far away from the rest. It may be an **error** (typing 500 instead of 50) or a **genuine rare case** (a billionaire in an income survey). Never delete an outlier without first asking why it is there.

> **The IQR rule (1.5 × IQR) – a popular way to flag outliers**
> Using our marks: Q1 = 52.5, Q3 = 67.5, IQR = 15
> Lower fence = Q1 – 1.5 × IQR = 52.5 – 22.5 = **30**
> Upper fence = Q3 + 1.5 × IQR = 67.5 + 22.5 = **90**
> Any mark below 30 or above 90 is flagged. If a new student scored 100, we would investigate it: a genius, or a data-entry mistake?

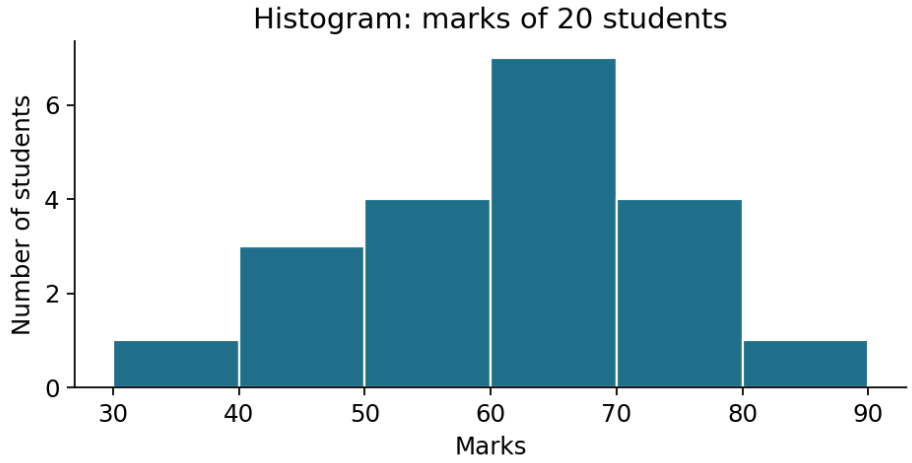
Another common approach is the **Z-score**: it tells how many standard deviations a value is from the mean. Values more than about 3 standard deviations away are usually treated as outliers.

6. Visual EDA – Popular Charts

'A picture is worth a thousand numbers.' Charts reveal shape, patterns and outliers that tables hide.

6.1 Histogram – shape of ONE numerical variable

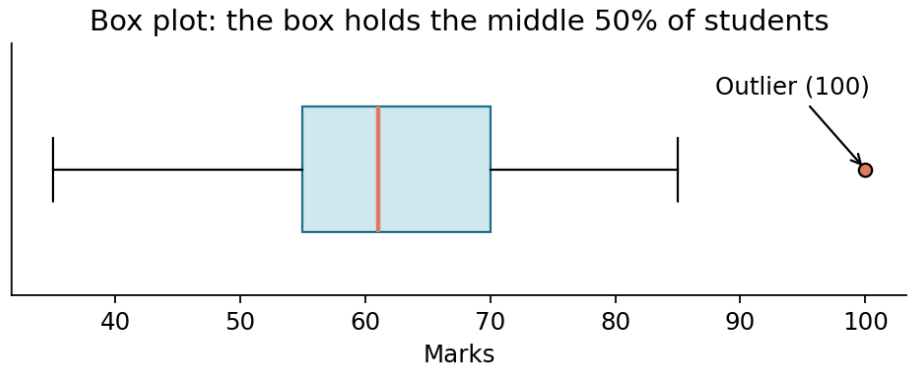
Values are grouped into ranges ('bins') and bars show how many fall in each range. Use it to see where most values lie and whether the data is skewed.



Reading it: most students scored between 50 and 70; very few scored below 40 or above 80.

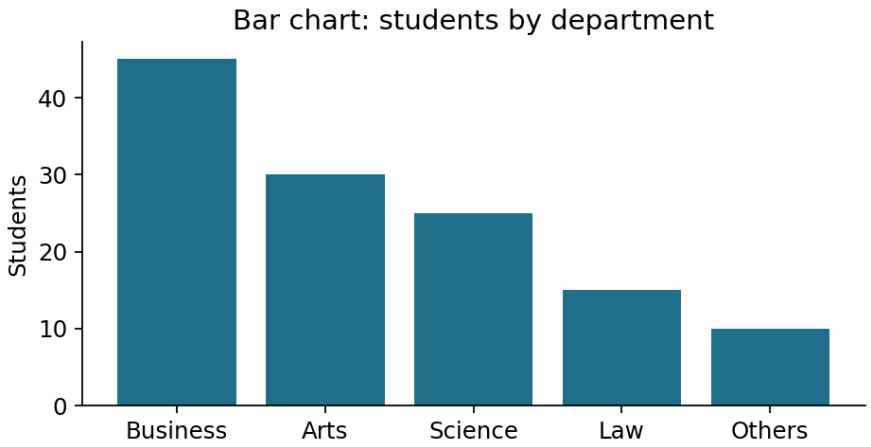
6.2 Box plot – summary and outliers at a glance

The box runs from Q1 to Q3, the line inside is the median, the 'whiskers' reach the normal range, and dots beyond are outliers. Great for comparing groups side by side (e.g. marks of Business vs. Science).



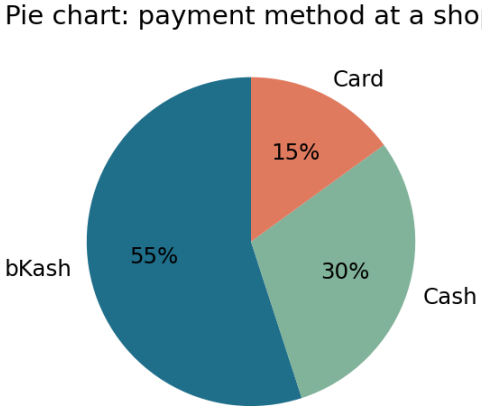
6.3 Bar chart – comparing categories

One bar per category; its height shows the count or total. Use for department, district, product type.



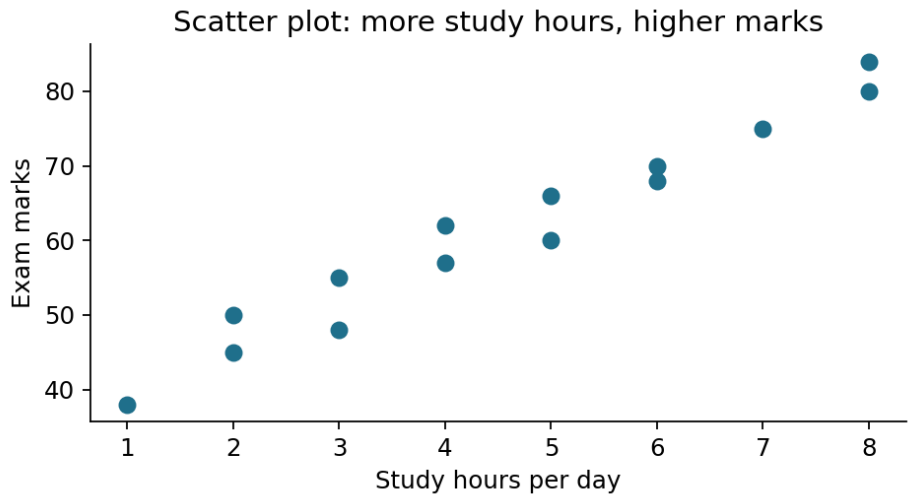
6.4 Pie chart – parts of a whole

Shows each category's share of 100%. Best with only a few categories (ideally 5 or fewer); otherwise use a bar chart.



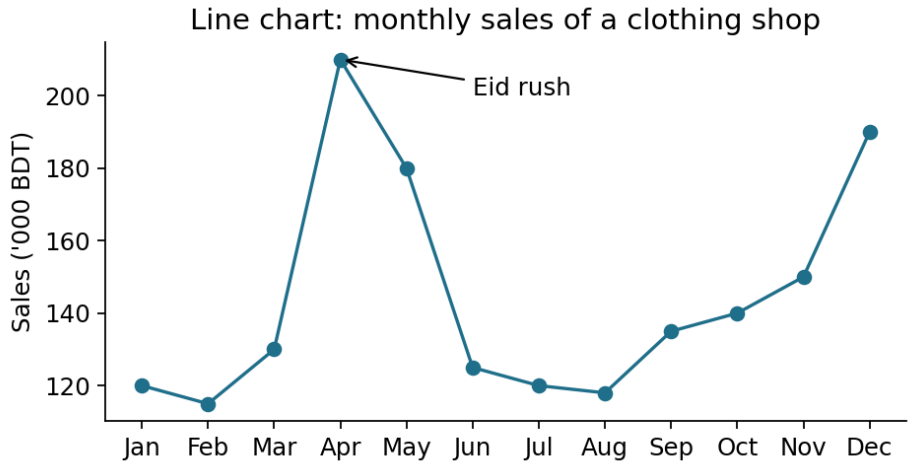
6.5 Scatter plot – relationship between TWO numerical variables

Each dot is one observation. If dots rise from left to right, the variables move together (positive relationship).



6.6 Line chart – change over time

Connects values in time order. Use to spot **trends** (going up or down) and **seasonality** (repeating patterns).

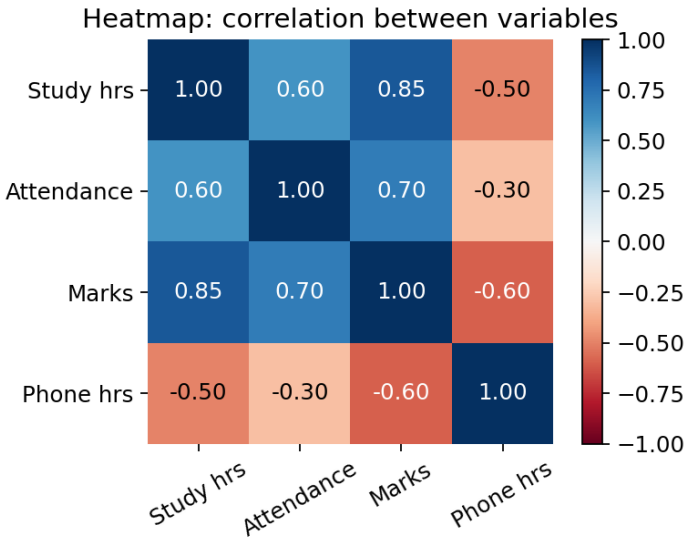


6.7 Correlation heatmap – many relationships at once

Correlation is a number from –1 to +1 describing how two numerical variables move together:

- **+1**: perfectly move together (more study hours, more marks).
- **0**: no linear relationship (shoe size and marks).
- **–1**: perfectly move in opposite directions (more phone hours, fewer marks).

A heatmap colours each pair so strong relationships stand out immediately.



Which chart should I use?

Your goal	Data type	Best chart
See the shape / distribution	1 numerical	Histogram, box plot
Find outliers	1 numerical	Box plot
Compare categories	1 categorical	Bar chart
Show share of a whole	1 categorical (few groups)	Pie chart
Relationship between two numbers	2 numerical	Scatter plot
Compare a number across groups	1 numerical + 1 categorical	Side-by-side box plots, grouped bar chart
Relationship between two categories	2 categorical	Cross-table (contingency table), stacked bar chart
Trend over time	Numerical + date	Line chart
Many relationships together	Many numerical	Correlation heatmap

7. Univariate, Bivariate and Multivariate Analysis

Type	Looks at	Example question	Typical tools
Univariate	One variable	What is the average age of our customers?	Mean, median, histogram, bar chart
Bivariate	Two variables	Do customers with higher income spend more?	Scatter plot, correlation, cross-table
Multivariate	Three or more	How do age, income and city together affect spending?	Heatmap, colour-coded scatter plot

Cross-table example (two categorical variables)
Survey of 100 customers: of 60 who live in Dhaka, 45 use mobile banking (75%); of 40 outside Dhaka, 16 use it (40%). Insight: mobile-banking use is much higher in Dhaka – a clue for a marketing campaign outside the capital.

8. Common Mistakes to Avoid

Correlation is NOT causation
Ice-cream sales and drowning cases both rise in summer. Ice cream does not cause drowning – **hot weather** (a hidden third factor) drives both. A relationship in data is a clue, not proof.

- Relying only on the mean when the data has extreme values.
- Deleting outliers without investigating them.
- Ignoring missing values or filling them carelessly.
- Using a pie chart with too many slices, or 3-D charts that distort sizes.
- Starting a bar chart's axis above zero, which exaggerates small differences.
- Looking only at numbers – always also **plot** the data. (Datasets with identical averages can look completely different when drawn; a famous demonstration is **Anscombe's Quartet**.)
- Forgetting the business context – always ask 'does this make sense in the real world?'

9. EDA Checklist

✓	Question
☐	Do I know what each column means and its type?
☐	How many rows and columns are there?
☐	Are there missing values? How will I handle them?
☐	Are there duplicates or inconsistent spellings?
☐	Are there impossible values or outliers? Why are they there?
☐	What are the mean, median, min, max and spread?
☐	Have I drawn the distribution of the key variables?
☐	Which variables are related to each other?
☐	Have I written down my key insights and next questions?

10. Quick Glossary

Term	Simple meaning
Variable	A column; one characteristic being measured
Observation	A row; one item studied
Distribution	How values are spread across possible values
Outlier	A value unusually far from the others
Skewness	Lopsidedness of a distribution
Quartile	Value splitting sorted data into quarters
IQR	Q3 – Q1; spread of the middle half
Correlation	Strength and direction of a relationship (–1 to +1)
Imputation	Filling in missing values
Insight	A useful finding that can guide a decision

11. Practice Questions

1. Daily sales (thousand BDT) of a tea stall for 7 days: 3, 4, 4, 5, 5, 5, 20. Find the mean, median and mode. Which best describes a typical day, and why?
2. For the marks 40, 50, 55, 60, 60, 65, 70, 80 (Q1 = 52.5, Q3 = 67.5), would a score of 25 be flagged as an outlier by the IQR rule? Show your working.
3. Which chart would you use for: (a) the number of customers in each district, (b) monthly electricity bills over two years, (c) the link between advertising spend and sales?
4. Classify each variable: blood group, number of siblings, monthly income, customer rating (poor / fair / good).
5. A study finds that people who own more books have higher incomes. Can we say buying books makes people richer? Explain.
6. A dataset lists a customer's age as 250. Give two possible reasons and describe what you would do.

> **Answer hints**
> 1) Mean ≈ 6.57, median = 5, mode = 5 – the median/mode, since the day with 20 is extreme. 2) Lower fence is 30; 25 < 30, so yes. 3) Bar chart, line chart, scatter plot. 4) Nominal, discrete, continuous, ordinal. 5) No – correlation is not causation (education or family background may drive both). 6) Typing error (25) or a wrong-column entry; check the source, correct it, or treat it as missing.